



Review Article

Polycystic ovarian syndrome, aetiology, pathophysiology, current management and future therapeutics with emphasis on adolescent girls

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Abstract

Polycystic ovary syndrome (PCOS) is a condition which affects the reproductive age women and involves hormonal changes. Generally, it could be seen in adolescents with a fluctuating symptom over a period of time. Women during reproductive age is affected with (PCOS) is about 5-20%. 70% of the women affected is not diagnosed worldwide. Infertility is one of the conditions for which Polycystic ovary syndrome (PCOS) is responsible. This is an incurable and chronic condition. It can cause imbalances in hormone, irregular menstruation, high androgen level and cyst in the ovary. Lack of ovulation will be the reason for irregular periods and it can cause difficult for the women to get pregnant. Despite through medications, life style modification and treatment for fertility symptoms may be improved. Women with family history of type 2 diabetes are found to be at higher risk for or type 2 diabetes are at higher risk. (WHO, 2023). The Rotterdam criteria for women are outlined to diagnose and confirm polycystic ovarian syndrome. Clinical signs of excess androgen, oligomenorrhoea or amenorrhoea are the features of PCOS, can be regarded as being at risk for PCOS. The disorder PCOS clinically is a complex condition with lifelong complications. This condition is more prevailing among reproductive age women. The characteristics symptoms and the diagnostic criteria on time are the most challenging aspect. Timely implementation and carrying out of PCOS management will reduce overall associated conditions and improve the quality of life. Further research studies are required on genetics and pathophysiology of PCOS for identifying and planning better strategies and therapeutic approaches. In order to delay the consequential long-term effect of the disease, significant effort should be taken on to investigate and improve the treatment measures. Various emerging therapies may have direct benefit in the management of the metabolic aspect of the metabolic aspect of PCOS; despite that clinical studies are required to examine its effectiveness as well as safety of PCOS women.

Keywords: Polycystic ovarian syndrome, Aetiology, Pathophysiology, Future therapeutics.

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1. Introduction

Married woman with infertility presented with pigeon size eggs with shiny surface was described and found by an Italian physician, scientist at 1721.

Sclerotic, fibrous, degenerative hydrops follicle of ovaries were described by Chereau and Rokitansky in 1844.

Bilateral oophorectomy was proposed by Lawson Tait in 1879.

Pathology and clinical implications of ovarian review was published by Von Kahlden in 1902.

The puncturing of cysts on the ovarian surface was suggested by John A. McGlinn in 1915.

In 1935 Stein and Leventhal introduced seven females found with the features of disturbances in menstruation, increased size of ovaries with marked presence of many tiny follicles.¹ Those women described with increased ovarian volume and lack of menstruation were suggested for ovarian wedge resection.²

The 1980s marked a turning point with the development of hormonal assays that identified hyperandrogenism as a

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central feature. Subsequent international consensus meetings, such as the NIH (1990), Rotterdam (2003), and AE-PCOS (2006) criteria, refined the diagnostic definitions and highlighted the heterogeneity of the syndrome. More recent research has expanded its scope to include genetic, metabolic, and environmental contributors, reflecting the shift toward understanding PCOS as a multifactorial, lifelong condition rather than one confined to reproductive health.³

2. Definition

A hormonal disorder commonly seen in women characterized by menstrual irregularities; increased blood level male sex hormone, excessive hair growth in different parts of the body and acne known as polycystic ovarian syndrome. (WHO, 2023).

Polycystic Ovarian Syndrome (PCOS) is a heterogeneous endocrine and metabolic disorder characterized by ovarian dysfunction, hyperandrogenism, and polycystic ovarian morphology. It affects women of reproductive age and often manifests as menstrual irregularities, infertility, acne, and hirsutism. According to the Rotterdam criteria (2003), PCOS is diagnosed when any two of the following three features are present, after excluding related disorders:

1. Oligo- or anovulation,
2. Clinical and/or biochemical signs of hyperandrogenism, and
3. Polycystic ovaries on ultrasound.

PCOS is now recognized as a lifelong condition extending beyond reproductive concerns, with implications for metabolic, cardiovascular, and psychological health. The syndrome's presentation may vary widely among individuals, especially during adolescence, necessitating age-appropriate diagnostic consideration.⁴

2.1. Causes

1. Insulin resistance: Excessive male hormone androgen will result in too much insulin resistance that causes disturbance in ovulation. There will be velvety dark patches in skin especially seen at neck, underarm, under the breast and groin area. These signs are suggestive of resistance insulin.⁵
2. Low-grade inflammation: PCOS patient will be seen with low grade inflammation which further cause the ovaries to make more hormone androgen.
3. Heredity: Mutations in CYP11A1, CYP11B2, CYP17A1, CYP19A1, CYP1A1, CYP21A2, CYP3A7 genes play a major role in pathological changes in PCOS that results in excessive blood levels of androgen and ovary dysfunction.
4. Surplus androgen: Excessive androgen interferes with ovulation. Due to which the egg won't be released on time also results in acne and hirsutism in the various body parts.⁶

2.2. Risk factors

1. Unhealthy behaviors arise in increasing weight, obesity, resistance to insulin, hyperinsulinemia and hyperandrogenism.
2. Excessive use up of drinks or diet items carrying rich in fructose, sugar, trans fat, fat from animals and processed foods.
3. Physical inactivity, stress and obesity.
4. Polycystic ovarian syndrome (PCOS) arises from a multifactorial interplay of genetic, environmental, and lifestyle influences.
5. Genetic predisposition plays a major role, with familial clustering and twin studies suggesting heritability rates up to 70%. Variants in genes related to androgen biosynthesis, insulin signalling, and gonadotropin regulation have been implicated.
6. Obesity, particularly central adiposity, exacerbates insulin resistance and hormonal imbalance, increasing disease severity.
7. Lifestyle factors such as sedentary behaviour, poor dietary patterns, and psychosocial stress further elevate risk, especially during adolescence when hormonal fluctuations are naturally pronounced.
8. Environmental exposures, including endocrine-disrupting chemicals like bisphenol A (BPA), have also been linked to PCOS-like features through their impact on steroidogenesis and metabolic regulation.
9. Early life factors—such as low birth weight, premature adrenarche, or intrauterine androgen exposure—may predispose girls to develop PCOS later in life.⁷

3. Discussion

3.1. Pathophysiology of PCOS

3.1.1. Ovulatory mechanism in normal woman

In normally ovulatory women, in response to properly functioning hypothalamic GnRH stimulation

Two important hormones produced from anterior portion of pituitary gland.

Follicular Stimulating hormone and Luteinizing hormone

Maturation and growth of ovarian follicles occur due to Follicular Stimulating hormone.

The follicle that obtains receptors of FSH will be dominant, grow and mature continuously on that month.

The dominant follicle maturation causes production of oestradiol hormone

When there is an excessive production of oestrogen that signals Follicular Stimulating hormone to stop production via negative feedback system.

Sustained and excess level of oestrogen will stimulate onetime rise of LH lead ovulation.⁹

3.1.2. Ovulatory mechanism in pcos woman

In PCOS woman, Hypothalamic GnRH is altered.

That results in increased Luteinizing hormone production from pituitary gland. The Luteinizing hormone causes theca stimulation results in production of androstenedione and testosterone. The excess androgen milieu of the ovary make it impossible for spontaneous growth, maturation and ovulation of follicles. Due to which the ovary becomes collected with so many tiny antral follicles which become not at all grow and dominant. This results in increased ovarian size and basal serum oestrogen.⁸

3.2. Symptoms

1. Heavy, long, intermittent, unpredictable or absent periods
2. Infertility
3. Skin will be oily, presence of acne
4. Growth of hairs in body and face
5. Thinning of hairs and baldness
6. Increased Weight
7. Acanthosis nigricans which is darkening of the skin which looks like purple or black in colour in the underarm, collar areas and nape of the neck, inner thighs, around lips, under eyes and forehead.¹⁰

3.3. Diagnosis

1. History collection includes menstrual history and weight gain.
2. Physical exam to note the signs of excessive growth of hair in the body and acne.
3. Pelvic examination
4. Blood test for hormonal studies
5. Blood testing also includes fasting cholesterol and triglyceride levels.
6. A glucose tolerance test to measures body's glucose response.
7. Pelvic ultrasound
8. Depression and anxiety screening
9. Obstructive sleep apnoea screening

3.4. Complications

Complications of PCOS can include:

1. Infertility
2. High blood pressure
3. Diabetes mellitus
4. Chances of miscarriage or premature birth
5. Fatty liver and inflammation of liver
6. Cardiac and blood vessel disorders
7. Sleep apnoea
8. Depression
9. Anxiety

10. Eating disorders
11. Endometrial cancer¹¹

3.5. Treatment

3.5.1. Lifestyle changes

1. Low-calorie diet
2. Moderate exercise activities.

3.5.2. Medications

1. Combination birth control pills.
 - a. Pills that contain both estrogen and progestin. This lowers the risk of endometrial cancer and correct irregular bleeding, excess hair growth and acne.
2. Progestin therapy
3. Progestin for 10 to 14 days every 1 to 2 months will regulate menstruation and prevent endometrial cancer.¹²
4. Clomiphene: It is an anti-oestrogen drug.
5. Metformin: It will lower insulin levels and improve insulin resistance.
6. Gonadotropins: Birth control pills to treat androgen production.¹³
7. Spironolactone (Aldactone). This medication treats excessive hair growth and acne inhibit the reaction of androgen on the skin, including over hair growth and acne.
8. Eflornithine (Vaniqa). To slow down the facial hair growth.
9. Hair removal
 - a. Electrolysis
 - b. Laser hair removal¹⁴

4. Lifestyle and Home Remedies

Lifestyle modification is the cornerstone of managing Polycystic Ovarian Syndrome (PCOS). A balanced diet rich in whole grains, lean proteins, fruits, vegetables, and healthy fats helps regulate blood sugar and reduce insulin resistance. Limiting refined carbohydrates and processed foods supports weight control and hormonal balance. Regular physical activity—such as brisk walking, cycling, or yoga—improves insulin sensitivity, promotes ovulation, and enhances mood. Stress management through mindfulness, meditation, or adequate sleep is equally important, as stress hormones can worsen PCOS symptoms. Home remedies like cinnamon, fenugreek, flaxseeds, and spearmint tea may provide mild benefits in reducing insulin resistance and androgen levels when used alongside medical advice. Maintaining a healthy body weight, avoiding smoking, and following consistent sleep routines can further improve hormonal and metabolic outcomes. Overall, small, sustainable lifestyle changes are more effective than restrictive diets or short-term interventions in managing PCOS.¹⁵

1. Healthy weight: Weight reduction can bring down androgen and insulin which will reimpose ovulation.
2. Carbohydrates limitation: Overconsumption of carbohydrate diets label insulin levels to reach peak. Consuming complex carbohydrates raises blood sugar very slowly. Fruits, whole grains, vegetables, peas and cooked dry beans contain complex carbohydrates.
3. Being active
 - a. Regular exercise
 - b. Increasing physical activity¹⁶
4. Diet and PCOS
 - a. Foods to add
 - i. Fiber rich vegetable like broccoli
 - ii. Fish and lean protein
 - iii. Anti-inflammatory foods
 - b. Spices: turmeric and tomatoes.

Fiber rich diets help in managing insulin resistance by lessening digestion and lowering the effect of blood glucose level. This may be beneficial for people with PCOS.

Examples of fibre rich diets

1. Cruciferous vegetables
 - a. Broccoli
 - b. Cauliflower
 - c. Brussels sprouts
2. Greens: red leaf lettuce and arugula
3. Green and red peppers
4. Beans and lentils
5. Almonds
6. Berries
7. Sweet potatoes
8. Winter squash
9. Pumpkin

Anti-inflammatory foods include:

1. Tomatoes
2. Kale
3. Spinach
4. Almonds and walnuts
5. olive oil
6. Fruits, like blueberries and strawberries
7. Fatty fish high in omega-3 fatty acids, like salmon and sardines¹⁷

Food to limit

1. Sugary drinks and snacks
2. Inflammatory foods like processed and red meats.
3. Refined carbohydrates
 - a. White bread
 - b. Muffins
 - c. Pastries
 - d. Sugary desserts

- e. Anything made with white flour
- f. Corn syrup
4. Beverages like soda and juice,
5. Fries, margarine, and red or processed meats.

Emerging therapeutics

1. Statins: This group of drugs includes fluvastatin, simvastatin Statins, atorvastatin, pravastatin, rosuvastatin helps in decreasing sex hormone production improves dyslipidaemia inflammation, and production of androgen.
2. Glucagon-like peptide-1 (GLP-1) Agonist: These drugs are helpful in reducing weight, fatty content in liver and lipid profile.
3. Inositols: Inositol is used in regulating follicular stimulating hormone and thyroid stimulating hormone. It also regulates metabolism, ovulation and minimizes the features of hyperandrogenism.
4. MicroRNA (miRNA) therapy: Trial studies are undergoing to test MicroRNA effectiveness in overweight and metabolic disorders, which ultimately helps in dealing with PCOS.
5. Interleukin (IL)-22 therapy: Interleukin (IL)-22 useful in ovarian dysfunction, resistance insulin and infertility.
6. Probiotics and prebiotics: Probiotics and Prebiotic improve hormones, regulate indicators of hormone disturbances in metabolism of lipids due to PCOS.⁽¹⁹⁾

5. Future Perspectives

PCOS is more prevailing among reproductive age women. The characteristic symptoms and the diagnostic criteria on time are the most challenging aspect. Timely implementation and carrying out of PCOS management will reduce overall associated conditions and improve the quality of life.

The next decade should prioritise development of adolescent-specific biomarkers and diagnostic algorithms that distinguish normal pubertal physiology from early pathologic trajectories. Improved biomarkers — combining hormonal panels (e.g., androgen fractions, AMH profiles), metabolic measures (insulin sensitivity indices), and emerging molecular or genetic signatures — could enable earlier, more accurate risk stratification and permit interventions tailored to a girl's underlying pathophysiology rather than a one-size-fits-all approach. This aligns with recent guideline calls for research to refine adolescent diagnostic criteria and validate novel markers in longitudinal cohorts.¹⁹

Nutraceuticals and insulin-sensitising supplements (for example, myo- and D-chiro-inositol) remain attractive adjuncts because of favourable safety profiles and ease of use; recent systematic reviews indicate potential metabolic and ovulatory benefits, but heterogeneity in formulations and study quality limits definitive recommendations. Well-

designed adolescent trials could clarify effective doses, combinations with lifestyle therapy, and which clinical phenotypes respond best, offering lower-risk alternatives or complements to prescription medications.¹⁹

Beyond new drugs, the future of adolescent PCOS care must embed prevention and psychosocial support into routine practice: school- and community-based obesity prevention, family-centred lifestyle programmes, and integrated mental-health services are essential to reduce incidence and improve long-term outcomes. Research priorities include pragmatic trials of multimodal interventions, cost-effectiveness analyses, and implementation science to scale evidence-based programmes in diverse settings. International guideline panels have emphasised these translational and implementation research gaps — particularly the need for adolescent-centric evidence.

Further research studies are required on PCOS pathogeny as well as on genetics for establishing and planning better action and therapeutic approaches. In order to slow up consequential enduring PCOS, significant effort should be taken on to investigate and improve the treatment measures. Various emerging therapies benefit in treating PCOS metabolic features, despite that clinical studies are required to examine its effectiveness as well as safety in women with PCOS. Further analysis is needed to prove the latent emerging therapeutics positively.

6. Conclusion

Polycystic Ovarian Syndrome (PCOS) in adolescent girls represents a complex interplay of genetic, hormonal, metabolic, and environmental factors. Early recognition during adolescence is crucial, as this period provides a unique opportunity to modify long-term health risks such as infertility, obesity, type 2 diabetes, and cardiovascular disease. The cornerstone of management remains lifestyle modification, emphasizing healthy nutrition, physical activity, and psychological support. Pharmacological therapy should be individualized, with combined oral contraceptives and metformin being the most widely used and evidence-based options for cycle regulation and metabolic control.

Emerging therapies—such as GLP-1 receptor agonists, inositols, and novel metabolic agents—offer promise but require further validation, especially for adolescent use. A multidisciplinary approach that includes paediatric endocrinologists, gynaecologists, nutritionists, and mental health professionals can ensure comprehensive care. Future research should focus on adolescent-specific diagnostic markers, early preventive strategies, and safe, targeted therapeutics to improve both reproductive and metabolic outcomes.

In conclusion, managing PCOS in adolescence should go beyond symptom control—it should aim at empowering

young girls with knowledge, lifestyle skills, and confidence to maintain lifelong reproductive and metabolic health.

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None.

8. Conflict of Interest

None.

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